DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics Engineering) SEMESTER - 2 Winter 2025 (Supply.)

Course :Bachelor of Technology (Electronics Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1000ES206B - PROGRAMMING FOR PROBLEM SOLVING

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 The unit of electrical resistance is:
a) Ohm b) Watt c) Henry d) Farad

2 Which element stores energy in the form of an electric field?
a) Resistor b) Inductor c) Capacitor d) Transformer

3 RMS value of a sinusoidal current is equal to:
a) Peak value b) Peak value / √2 c) Average value d) Zero

4 Which machine converts mechanical energy into electrical energy?
a) Motor b) Transformer c) Generator d) Induction motor

5 The working principle of an induction motor is based on:
a) Electromagnetic induction b) Electrostatics
c) Ohm's law d) Coulomb's law

6 CRO is mainly used to measure:
a) Resistance b) Power
c) Frequency and waveform d) Energy

7 Moving coil instruments are:
a) AC instruments b) DC instruments
c) Both AC & DC d) None of these

8 Digital Storage Oscilloscope is mainly used to:
a) Measure DC voltage only b) Display waveform and frequency
c) Measure current only d) Measure resistance

9 The total resistance of three resistors 2 Ω, 3 Ω, and 6 Ω connected in series is:
a) 11 Ω b) 1 Ω c) 6 Ω d) 3 Ω

10 Kirchhoff's Current Law (KCL) is based on:
a) Conservation of energy b) Conservation of charge
c) Ohm's law d) Lenz's law

11 Power factor in a purely resistive AC circuit is:
a) 0 b) 0.5 c) 1 d) -1

12 A Series DC motor is suitable for:
a) Constant speed applications b) High starting torque applications
c) Precision speed control d) Low power applications

Q2. Solve the following.

A)	Explain the Kirchhoff Current Law with a suitable example.	6
B)	Derive an expression for the average and the RMS value in an AC circuit.	6

Q3. Solve the following.

A)	Distinguished between Motors and Generators with examples.	6
B)	Derive the EMF equation for a transformer with a neatly labelled diagram.	6

Q4. Solve Any Two of the following.

A)	Explain the principle operation of a PN Junction Diode with a suitable diagram.	6
B)	Explain the working of a full-wave bridge rectifier with waveforms.	6
C)	Explain the functional block diagram of a DC power supply.	6

Q5. Solve Any Two of the following.

A)	Derive the relation between current gains in Common Base (α) and Common Emitter (β) configurations.	6
B)	Explain the input and output characteristics of a BJT in the Common Emitter(CE) configuration.	6
C)	Enlist Different Transistor Biasing Techniques, and explain the Fixed Biased Technique in detail.	6

Q6. Solve Any Two of the following.

A)	Explain the block Diagram of a function Generator.	6
B)	Describe the working principle of the Moving Coil instruments with a suitable diagram.	6
C)	Explain the operation of a Digital Storage Oscilloscope with suitable Diagram.	6

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